Rohit Kottomtharayil

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Education

Carnegie Mellon University **May 2028**

Bachelor of Science in Electrical and Computer Engineering — Dean's List *Pittsburgh, PA*

- Relevant Coursework:(15-122) Principles of Imperative Computation, Intro to Electrical & Computer Engineering, Concepts of Mathematics, (18-213) Introduction to Computer Systems, Differential Equations, (18-240) Structure and Design of Digital Systems
- Leadership & Activities: **Mayur South Asian Student Association** (President, 400+ members); **ScottyLabs** (Labrador Software Development Committee)

Experience

Nokia Bell Labs **June 2026 – August 2026**

Optics Hardware Intern *Murray Hill, NJ*

- Contributed to a collaborative prototype development project between Nokia Bell Labs and Google, specializing in FPGA design, PCB design, and GPU optimization for real-world Distributed Acoustic Sensing (DAS) systems.
- Characterized IR drop and thermal rise across a Nokia G34L midplane under staged DC loading with a four-channel electronic load to determine whether a bus bar was required for power integrity.
- Built a fleet telemetry platform (Python, Netmiko) that polls G34L optical shelves worldwide, stores health and inventory data in SQLite/InfluxDB, visualizes trends in Grafana, and alerts on-call staff via Teams.
- Automated u-blox ZED-F9T GNSS timing receiver bring-up with a Python CLI, and optimized flying-probe PCB test procedures to speed up board validation.

Build18 — Carnegie Mellon University **2025 – Present**

Sponsorships Chair *Pittsburgh, PA*

- Help organize Build18, CMU's entirely student-run, week-long hardware hackathon where 600+ students form 120 teams each January to design and build an electronics project, capped by a final project expo.
- Lead corporate sponsor outreach and relationships, plan and coordinate sponsor-hosted events, and secure the funding that provides each of the 120 teams with ∼$500 for parts and project builds.

Pfizer Inc. **September 2024 – January 2025**

Machine Learning Intern *New York, NY*

- Collaborated with two teammates to develop software access classification models successfully integrated into Pfizer workflows to streamline access to critical software and hardware.

Hack4Impact **June 2023 – June 2025**

High School President / Founder *Lincroft, NJ*

- Launched the first high school chapter of the nonprofit and led student teams building software for groups including the Hudson River Foundation and Nature Conservancy.

Projects

Fiber DAS Vehicle Detection | *DSP, Real-Time Systems, Edge Compute* Nokia Bell Labs

- Helped build a real-time vehicle detector turning buried fiber into vibration sensors (optical phase to directional counts), and profiled it from a lab GPU workstation down to edge Arm hardware.

Deluge — Real-Time Flood Simulation | *TypeScript, WebGPU* **2nd / 50+** — SteelHacks (MLH), No Wrapper Track

- Solved the 2-D shallow-water equations on the GPU with WebGPU compute shaders over real USGS elevation data, running live in the browser at 60 fps and ∼70× real time.
- Stabilized the solver (CFL-adaptive timestep, positivity-preserving flux limiter; mass error $\sim 10^{-7}$), validated it against NWS flood stages, and added levee building and live evacuation routing.

RV32IMC Pipelined RISC-V CPU (FPGA) | *Verilog, Verilator, cocotb*

- Designed a 5-stage RISC-V CPU with caches and branch prediction (2.88 CoreMark/MHz, 34.7 MHz on ECP5); verified via ISS co-simulation, 100% functional coverage, and 77 formal proofs.

Awards & Recognition

- **American Junior Academy of Science (AJAS)** — 1st Place at NJAS for environmental research; presented nationally.
- **Columbia Science Honors Program (SHP)** — Selective STEM program; advanced environmental coursework.
- **Classical Languages** — Latin Seal of Biliteracy; "Perfect Paper" on the National Latin Exam.

Technical Skills & Interests

Languages: Python, Java, C, TypeScript, MATLAB, SystemVerilog, Verilog, HTML/CSS, JavaScript
Hardware & Tools: FPGA Design, RTL Design, Static Timing Analysis, Formal Verification (SVA), PCB Design, Schematic Capture, High-Speed/Impedance-Controlled Routing, Signal & Power Integrity, Oscilloscopes, Logic Analyzers
Software & Frameworks: Git, KiCad, Vivado, Verilator, WebGPU, NumPy, Scikit-learn, TensorFlow, React, Node.js
Areas: Digital Design, Machine Learning, Embedded Systems, Optics Hardware, Signal Processing, Data Analysis